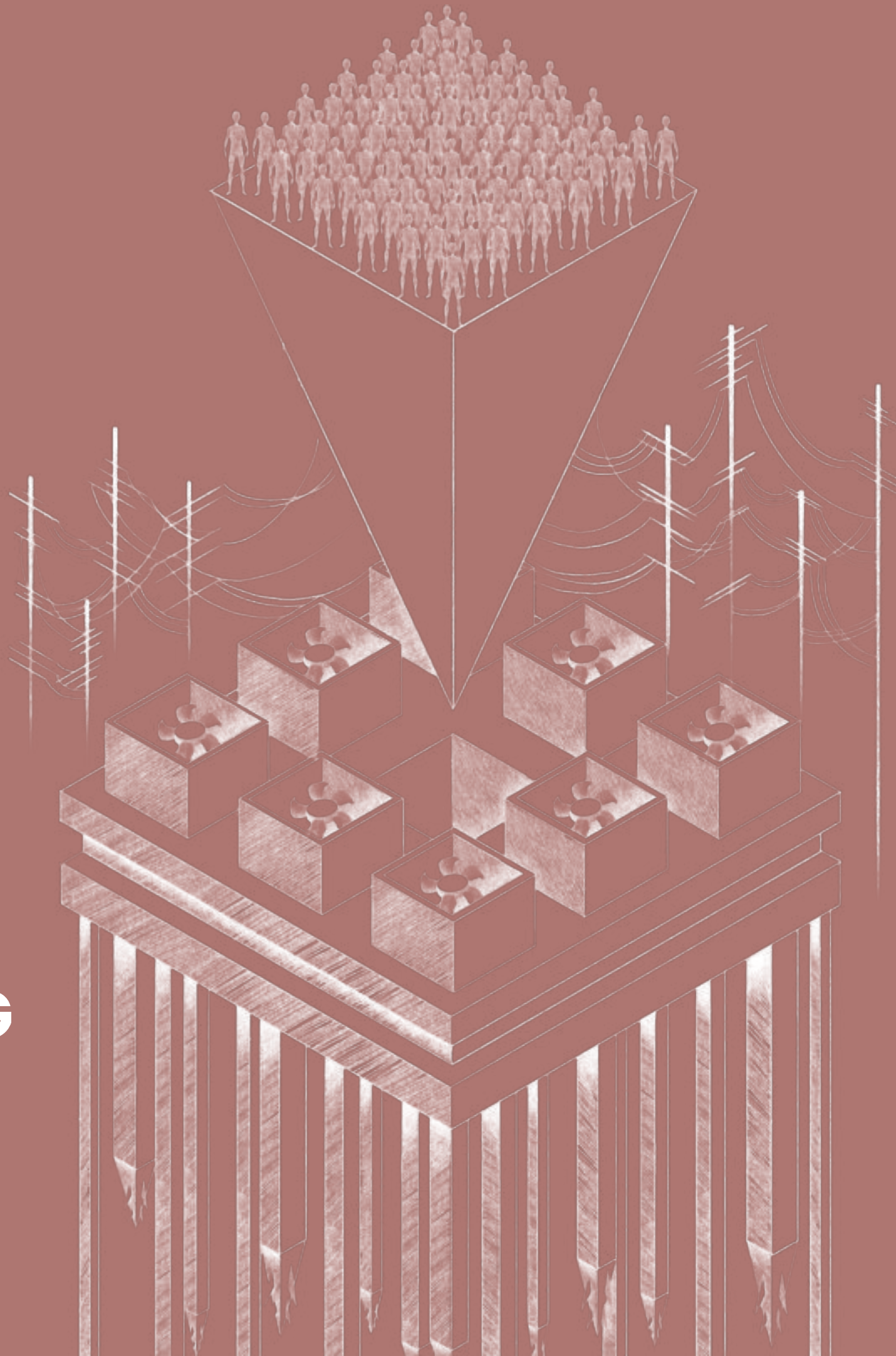


RETHINKING . LOGISTICS

REINTRODUCING
DATA CENTERS

AD 2

KATRIN RYBINSKIY 2026



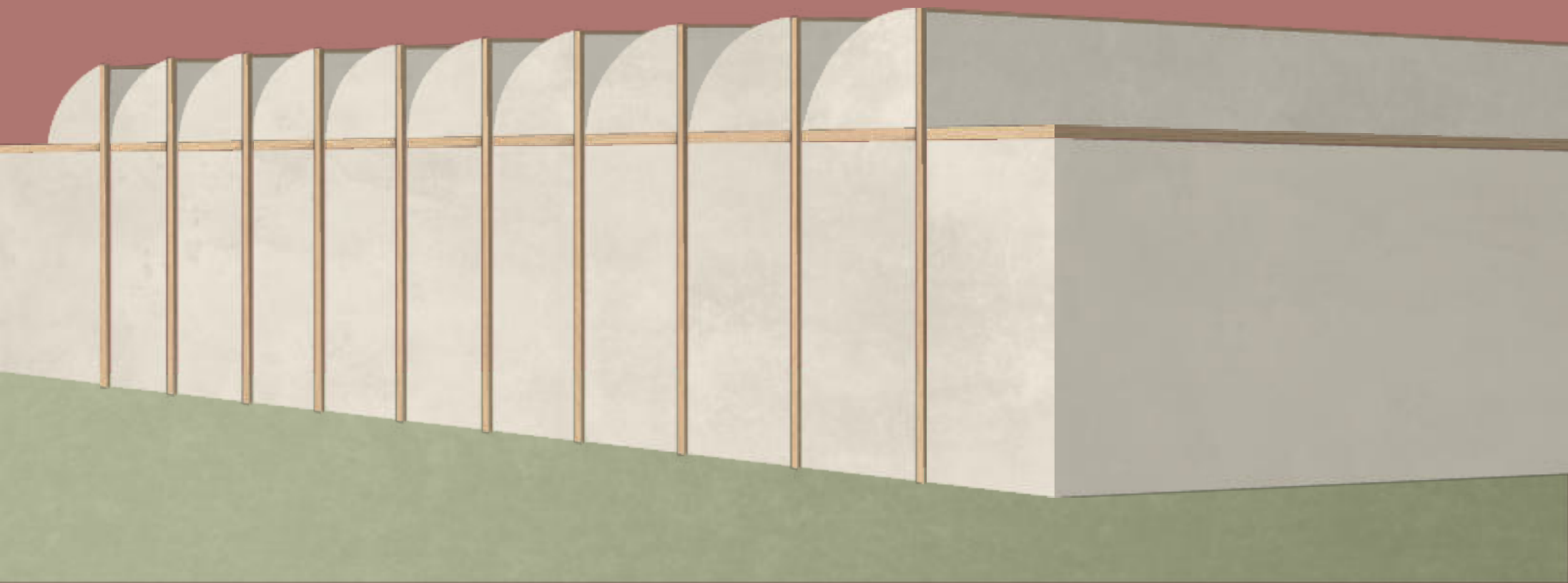
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INTRODUCTION TO STUDIO:

During this studio, we were introduced to multiple logistics typologies and needed to choose a topic to focus on. after some consideration I decided on data centers due to their rapidly growing popularity among investors, governments and big tech. I wanted to understand the issue and what can be better.

During the studio, we had the rare opportunity to receive presentations and feedback from people in the field of architecture, finance, LCA, Energy, landscape and more. they led us along the project and gave us precious input and data, as well as one on one attention in some cases. In addition, This semester, we worked with the double diamond system, which helped me organize my time effectively and gain a clearer understanding of what to focus on at each stage, whether it's research, concept or design.



What Is a Data Center?

STORAGE



More commonly known as cloud storage, this is a process that saves your data on remote servers hosted in large facilities called data centres, rather than on your own device.

When you "store something in the cloud," your data physically lives on hardware inside one of these buildings. Providers rarely store data in just one location; they use replication, copying your data across multiple drives, servers, and geographic regions so that if hardware fails, your data survives.

The primary storage hardware is SSDs (Solid State Drives), which support low-latency workloads and frequent access.

COMPUTING



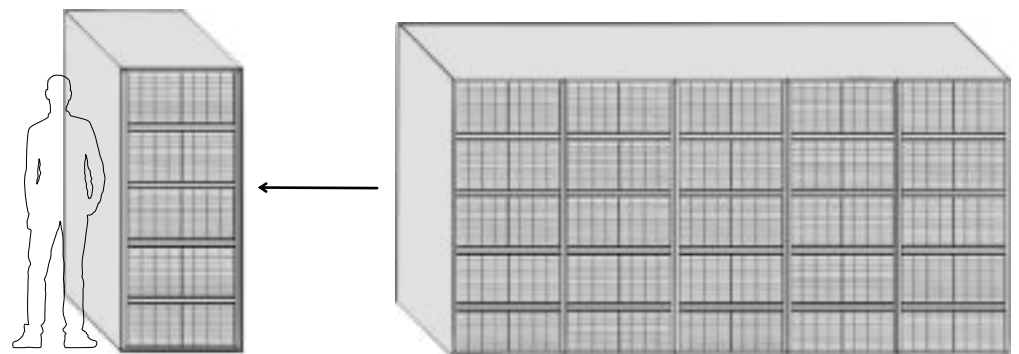
Computing data centres house large numbers of servers that process and distribute data at scale. They require significant energy and cooling infrastructure, as well as ongoing hardware upgrades.

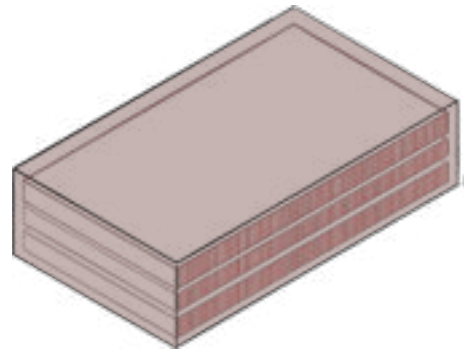
The most common cooling method is water-based. A 100 MW facility can consume 7-10 million litres of water per day. As AI workloads have grown sharply, so has water demand causing many operators to rely on open-loop evaporative cooling, which loses water to the atmosphere rather than recirculating it, placing pressure on local water supplies and the surrounding environment.

The sheer scale of energy demand also strains existing power grids. To meet grid connection requirements, data centres are increasingly built near cities, raising concerns about noise and visual impact on local communities.

SERVER RACK

Server racks are the physical steel frames that organize and house the servers inside a data centre. Each rack holds multiple servers stacked vertically, along with the networking equipment and power distribution units that connect them. Racks are arranged in rows, typically in alternating "hot" and "cold" aisles, a layout designed to direct cool air to the front of the servers and expel hot air from the back, improving cooling efficiency. A single rack can exceed 100 kW in high-density AI facilities, making power delivery and heat management the central engineering challenge of any modern data centre.



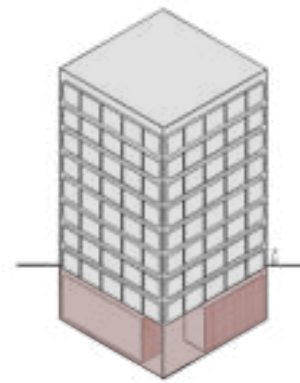


HYPERSCALE D.C

Massive structures operated by tech giants host thousands of servers and are primarily dedicated to computing operations.

size: 100,000m²,

energy consumption: 300–500mW ~270,000 average EU homes for a day

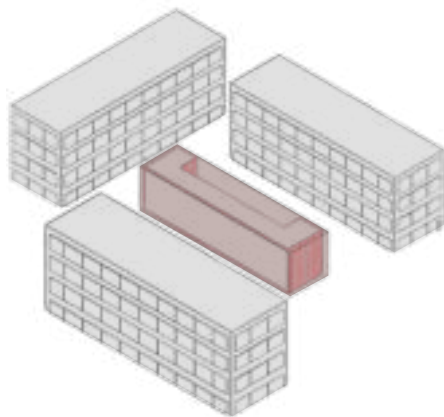


ENTERPRISE

Privately owned and operated by a single organization (bank, government...) to run their own IT infrastructure not for public cloud services. Usually located in the office building's basement.

Size: Dependent on office size
~500 – 2,000 m²

energy consumption: 2–3 MW

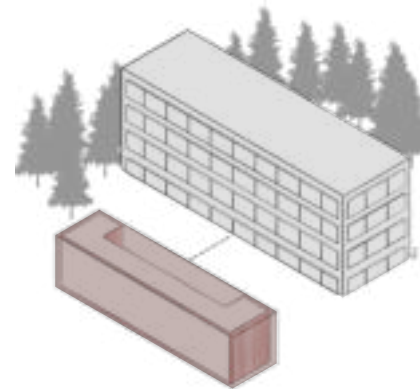


COLOCATION D.C

A third-party facility where multiple organizations rent space, power, and cooling, bringing their own servers and networking equipment. The colo provider owns the building and infrastructure; the tenant owns the hardware.

Size: 2,000–10,000 m²

Energy consumption: 5 MW – 30 MW

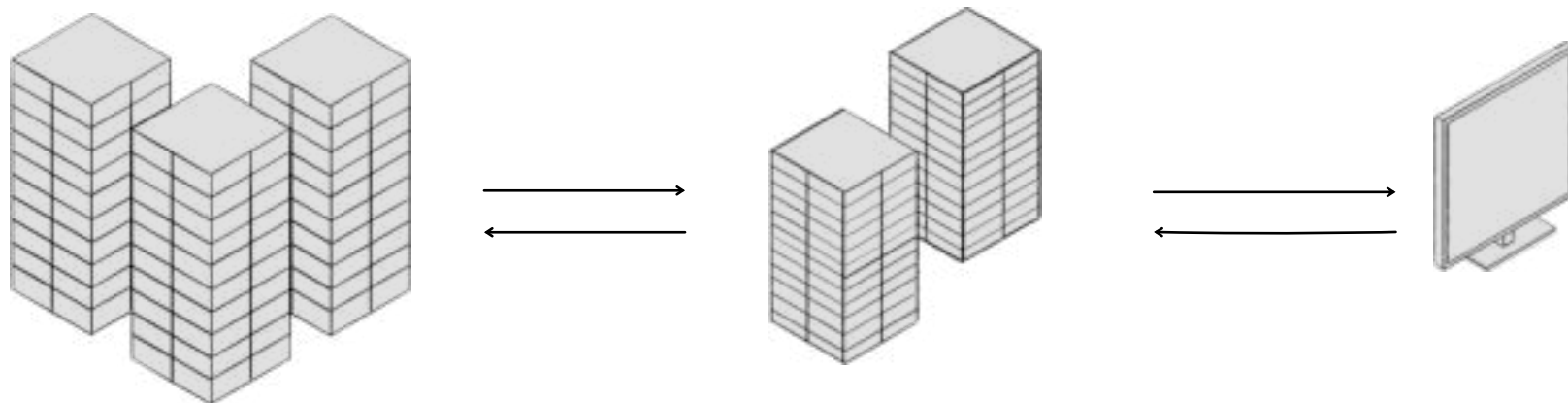


EDGE D.C

Small, distributed facilities located close to end users or data sources at the "edge" of the network rather than centralized in a core location. The goal is to minimize latency and reduce traffic to central data centers.

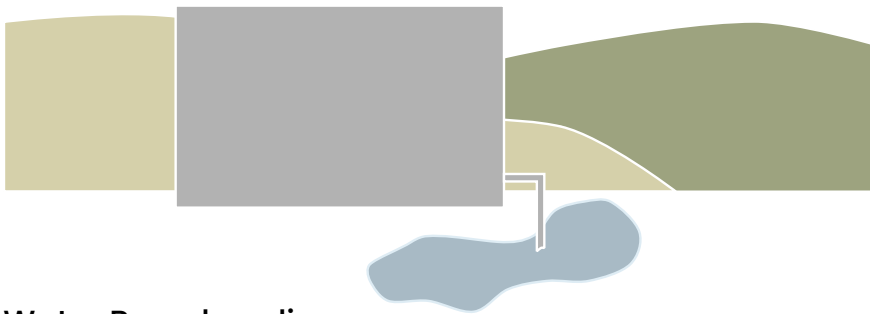
Size: 100–500 m²

Energy consumption: 100 kW – 1 MW



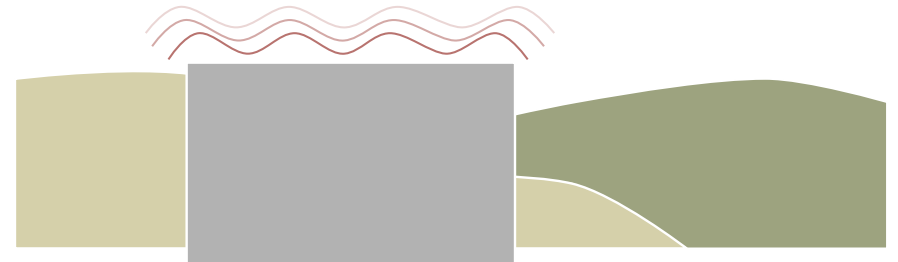
Most Data works like a web: data centres work with each other, transferring data from large data centres to smaller ones and then to your personal computer. They preserve information and, in turn, allow for low latency and quick access.

Why Are Data Centers Problematic?



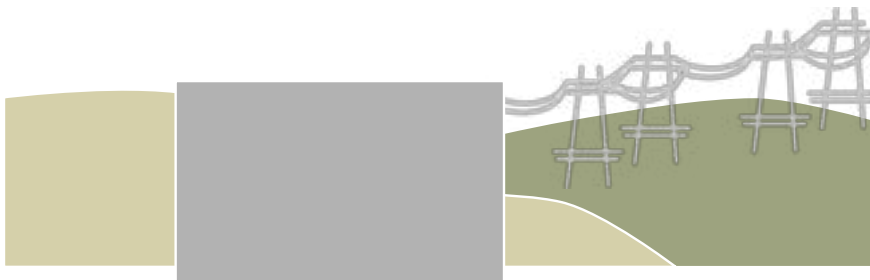
Water Based cooling

Some data centres pump water from underground to cool the immense heat the technology emits inside. Some, due to excessively fast construction, don't create a system that accumulates the vaporised water and just get rid of it. However, if a circular system is installed, the problem can be avoided by collecting the vaporised water and reusing it, thereby keeping water use minimal.



Excessive Heat Emissions

The area around data centres is, on average, 5 degrees higher within a 100-meter diameter, creating an unnatural temperature for the environment and exacerbating the already existing heat islands in cities.



Increasing energy demand

Data Centres are in rapidly increasing demand for energy, and some countries, like the Czech Republic, are struggling to accommodate this, and therefore, electricity prices increase for residents.



Built close to residential areas

Because the energy grid is more developed near cities, data centres (D.C) are often built extremely close to private homes. The D.C.s are built as very tall, large boxes that block residents' views and create problems such as noise, rising water levels, and higher electricity prices.

JOB DISPLACEMENT

Job displacement occurs when workers permanently lose their jobs due to structural changes in the economy rather than temporary downturns.

Unlike layoffs during a recession, displaced workers often can't return to their old roles even when the economy recovers — those jobs simply no longer exist.

Now, Job displacement is more visible than ever, as companies are turning to AI instead of human workers. Causing actively increasing unemployment rates and a plummeting workforce.

However, this is not a new narrative. Just 70 years ago, we had people working in manufacturing, but today they've mostly been replaced by more efficient machinery.

In 2025 alone, 55,000 jobs were directly attributed to AI substitutions.

Even in Creative Professions like architecture or art, AI has been exerting its influence and speeding up the workflow while leaving its mark on the “original” work.

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Even in Creative Professions like architecture or art, AI has been exerting its influence and speeding up the workflow while leaving its mark on the “original” work.

Nonetheless, there are positives. Hospitals have been using AI to search archives and identify diseases much more efficiently, and as a result, find cures and understand algorithms.

In conclusion, AI is here to stay, so we should be aware of the tasks we require of it and what lines we allow it to cross.

Concept:

My main idea is to reintroduce the concept of data centres by highlighting their design and appearance, as well as their effects on people, animals, and the environment.

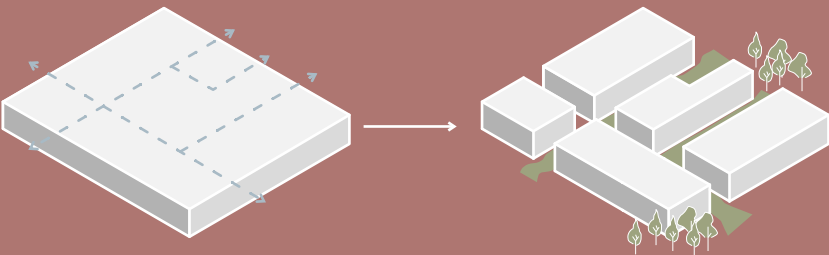
I focused on integrating data centres into the environment, rather than creating damaging buildings.

The areas I chose to emphasize are minimizing heat production, contributing positively to urban spaces, producing energy, and incorporating green and blue infrastructure.

I advanced by balancing areas allocated to natural elements and structures, incorporating renewable energy sources, and re-purposing existing buildings.

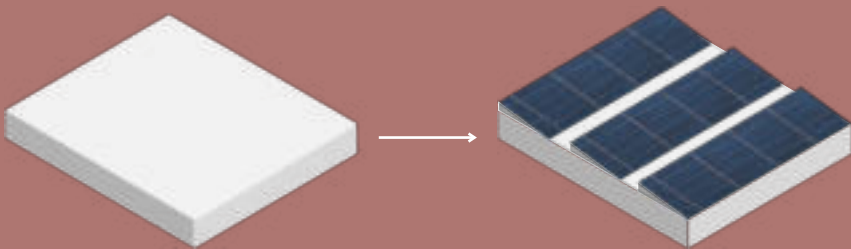
I approached it with careful consideration of the technical aspects and, wherever possible, with research as a fundamental rather than a secondary focus.

GREEN CORRIDORS



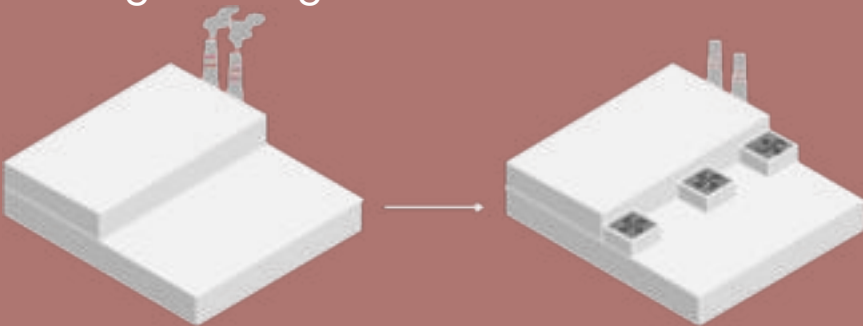
Creating more paths and allowing for green spaces that let small animals and people experience nature in an industrial sector.

GENERATING ENERGY



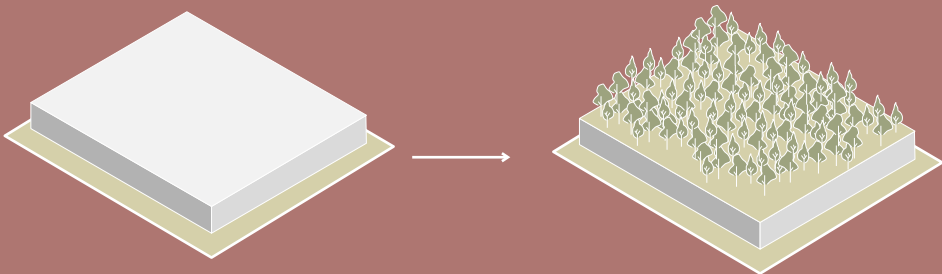
Green energy to maintain the site with minimal investment and the possibility of selling it to generate additional profit. As well as EV parking spots that can generate more profit if available to the public.

REUSING
existing buildings



Reusing existing structures, maintaining the city's industrial landscape and minimizing CO2 emissions

EQUALIZING
structures and green spaces



Reusing existing structures, maintaining the city's industrial landscape and minimizing CO2 emissions



INTRODUCTION TO SITE:

Kladno is a city in the Central Bohemian Region of the Czech Republic, located 25 km northwest of Prague. and forms part of the greater Prague metropolitan area.

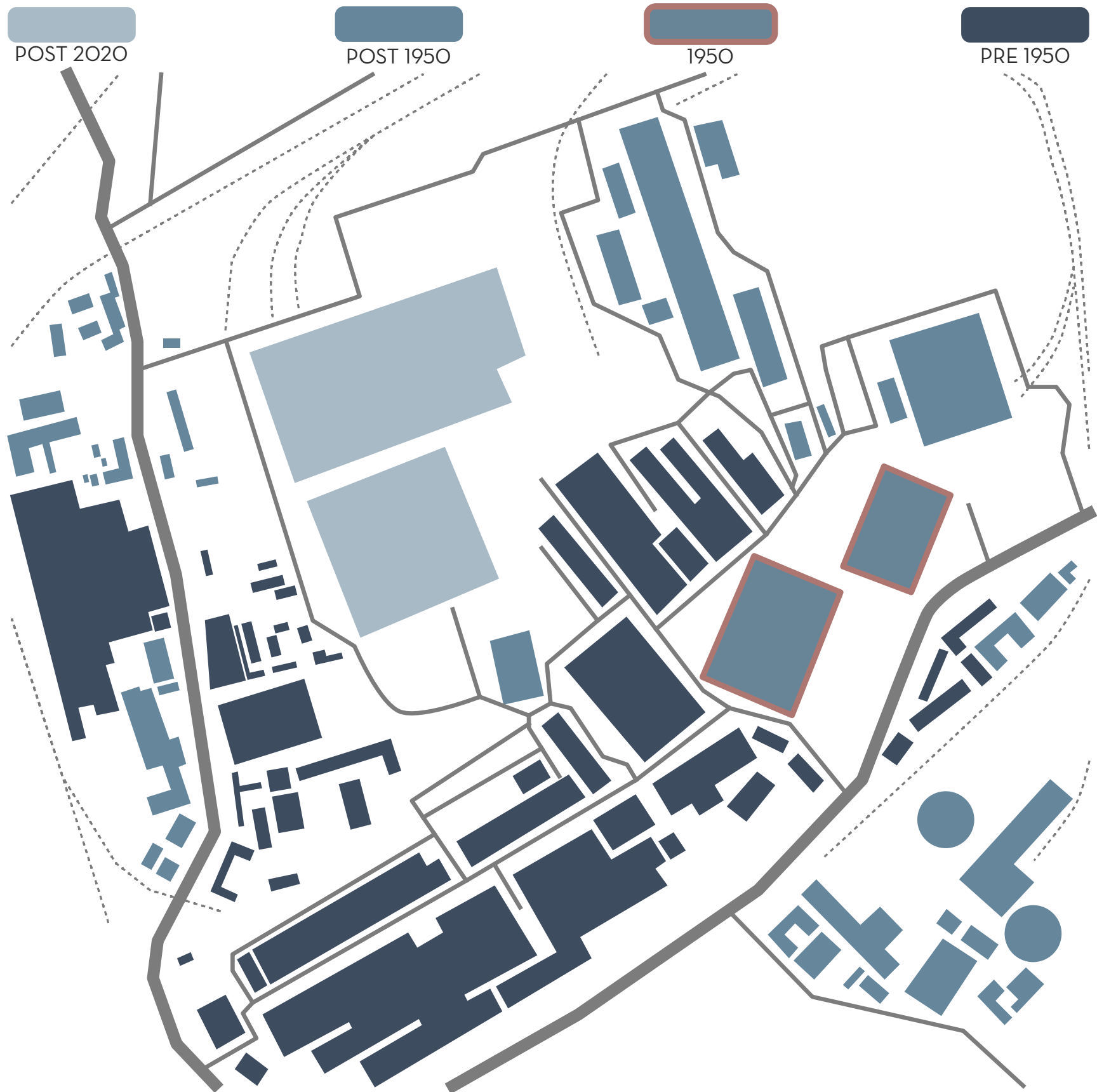
The first written evidence of Kladno dates to the 14th century. The growth of coal mining and the iron industry around 1850 led to a dramatic transformation from a small agricultural town into an important industrial region of Bohemia. In 1898, it was awarded the honorary title of royal mining town. The collapse of the Poldi steelworks after an unsuccessful privatization in the 1990s was the largest corporate bankruptcy in Czech history and hit the city hard, along with the closure of the coal mines.

Today, the city still preserves its industrial essence, and most residents still work in factories or warehouses. However, the mining industry in Kladno decreased, and open museums and public spaces took priority.

Many large companies see Kladno as a good location for investment due to its proximity to Prague and its hard-working residents.







Kladno industrial centre has lots of historical buildings that were primarily used as factories, we got the chance to see the abandoned building ourselves (outlined in red). It was a great inspiration to see spaces with such rich natural light and human-scale architecture. The spaces were clearly built to serve humans and not machines.



INDUSTRIAL



MINING



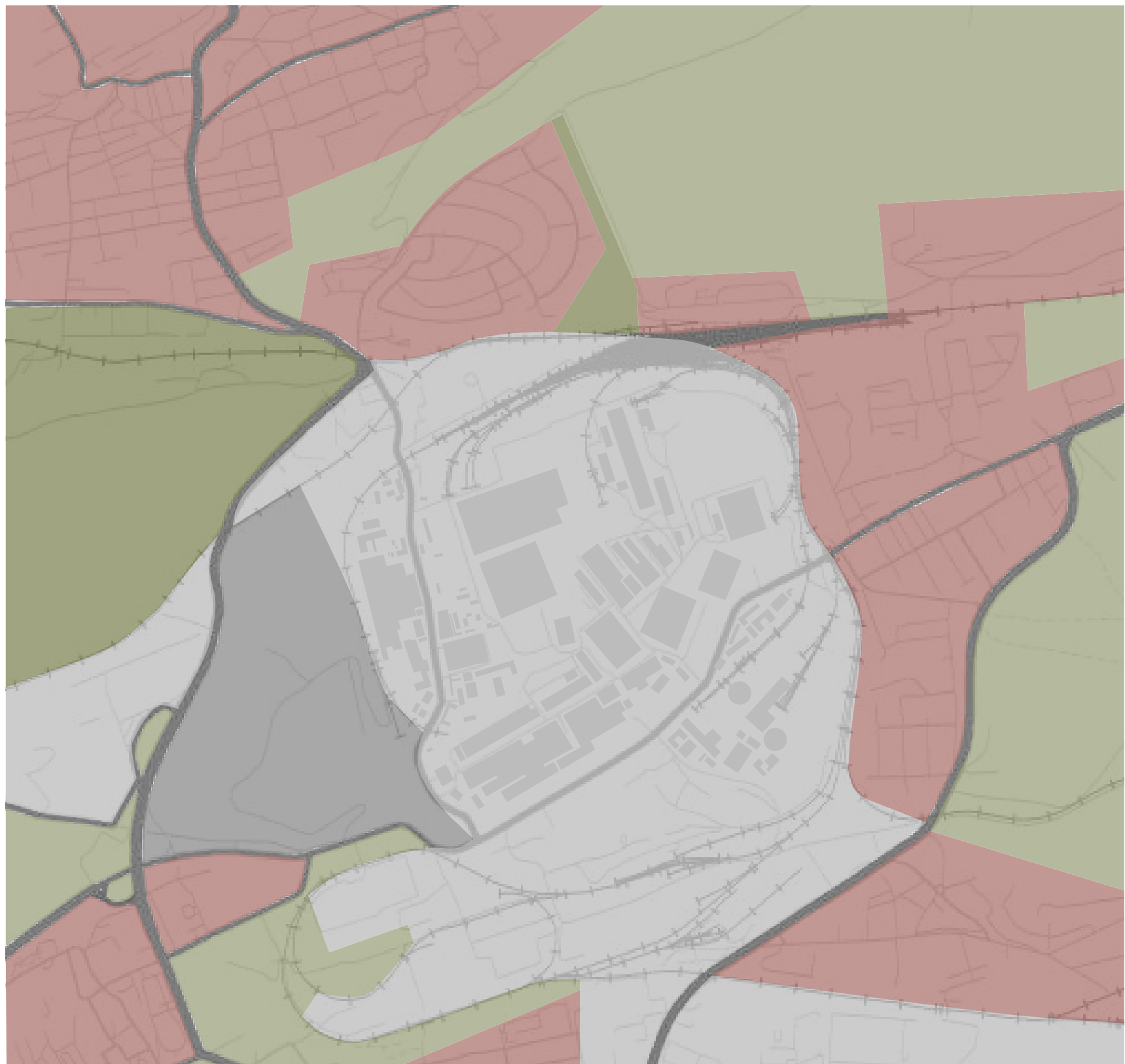
RESIDENTIAL



FOREST



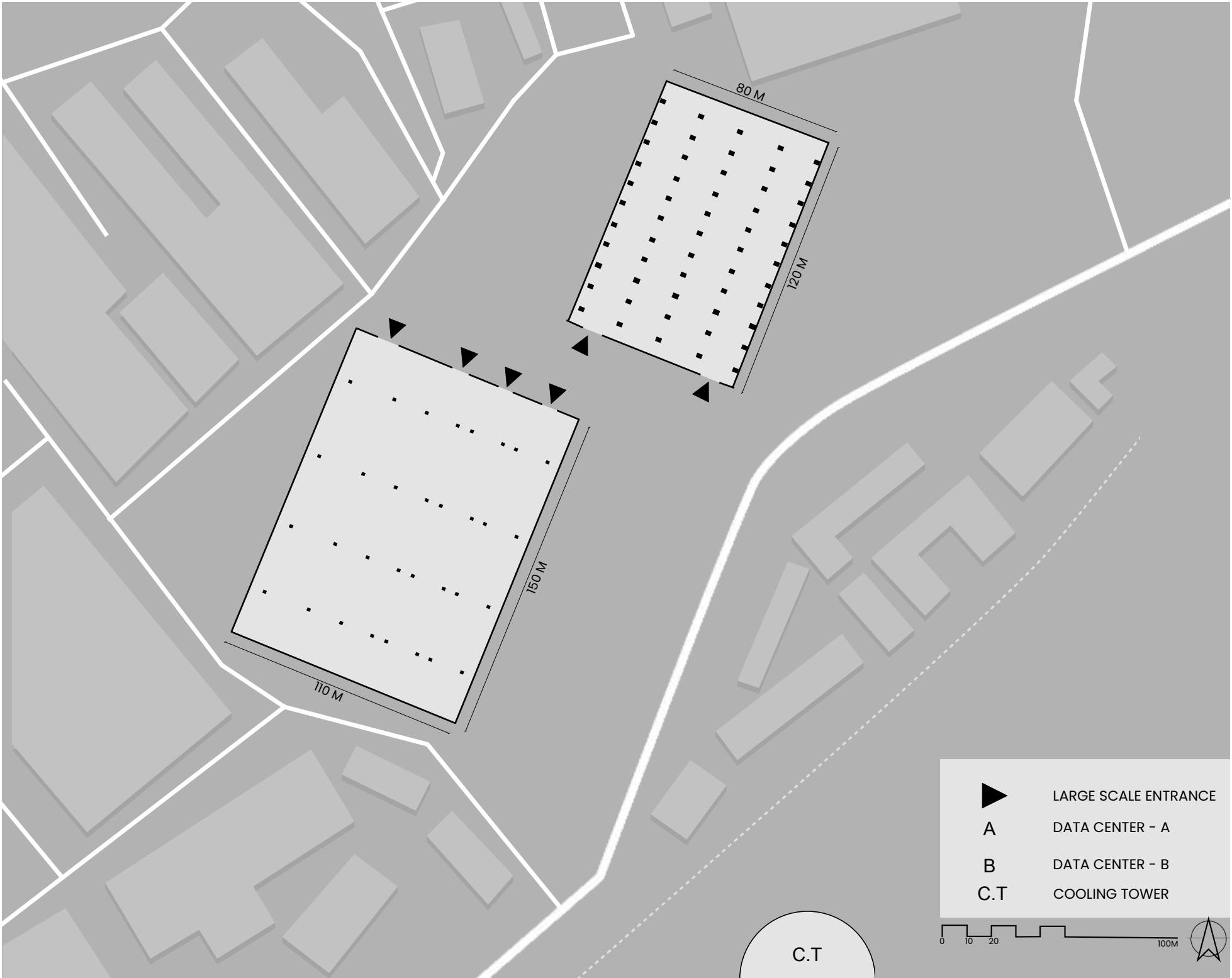
FIELD



The industrial centre of Kladno is surrounded by residential areas and forest, creating a unique dynamic between factories and residences. The industrial sector and the residential area are separated by a small valley with train tracks, creating a clear distinction between the two and allowing for a natural separation.

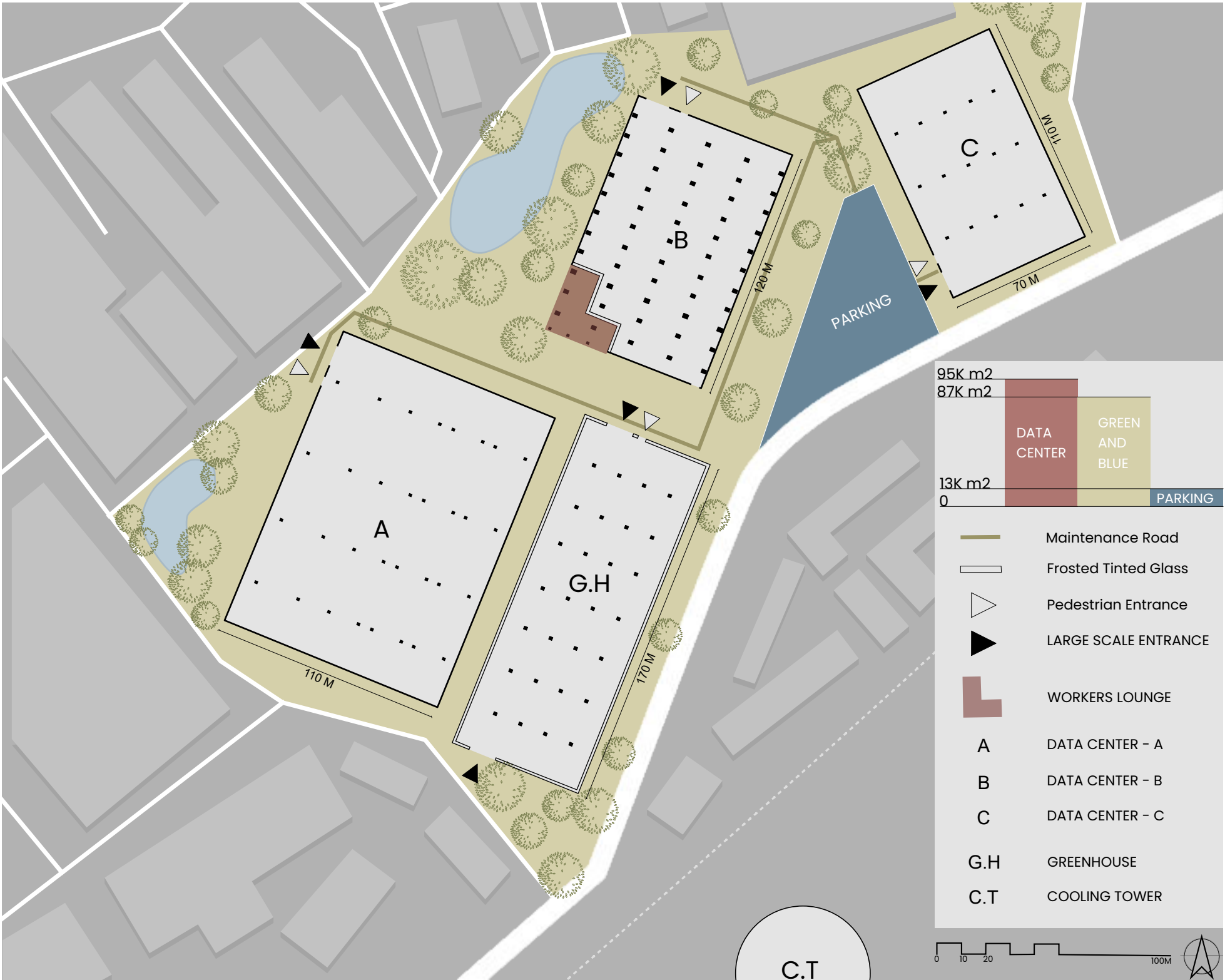
SITE BEFORE INTERVENTION

On the chosen site, there are two existing buildings built in the 1950s that used to serve as metal manufacturing factories. However, today they stand abandoned and unused, even though the building has good structural integrity and can serve for years more. In the last 5 years, the Plot was bought by a company, and the buildings are planned for demolition. However, I would like to preserve the original exterior and just adapt the interior for modern use.



SITE AFTER INTERVENTION

On the plan, the original two buildings are preserved, but building B was carved to create a long corridor for workers, with windows overlooking the data centre's interior. I added two more buildings, including building C, which will serve as a data centre with a roof covered in PV panels. As well as a greenhouse that will reuse the heat emitted by the data centres. In addition, I tried to maximize the green areas and minimize the pavement.



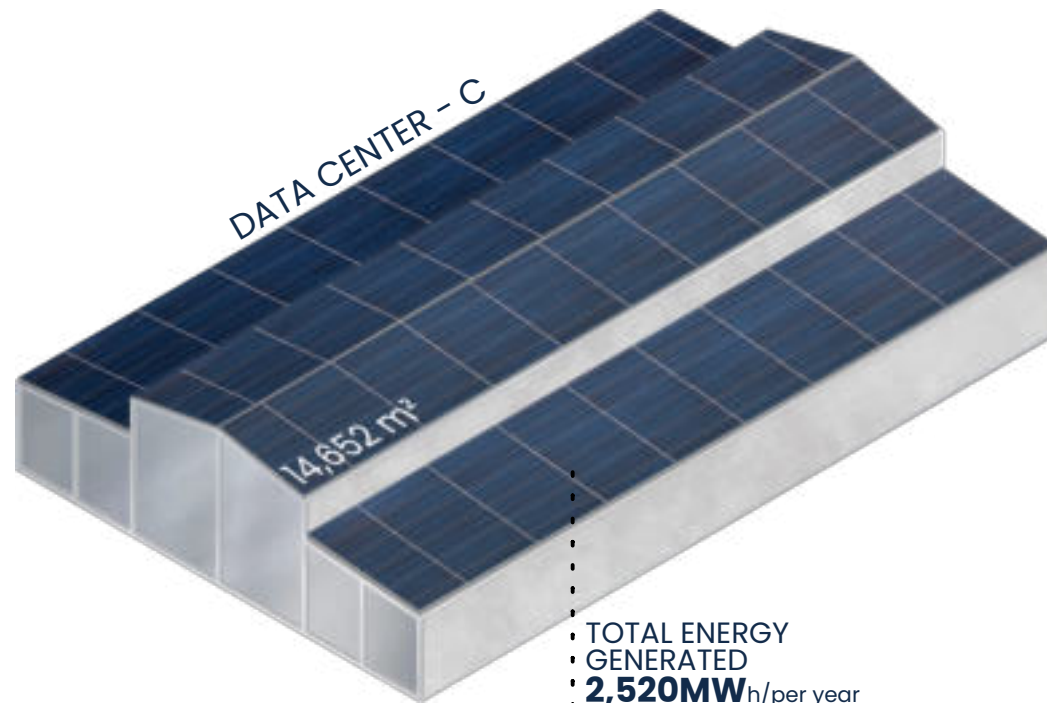
SOLAR ENERGY GENERATION:

Energy can be a source of benefit for data centres.

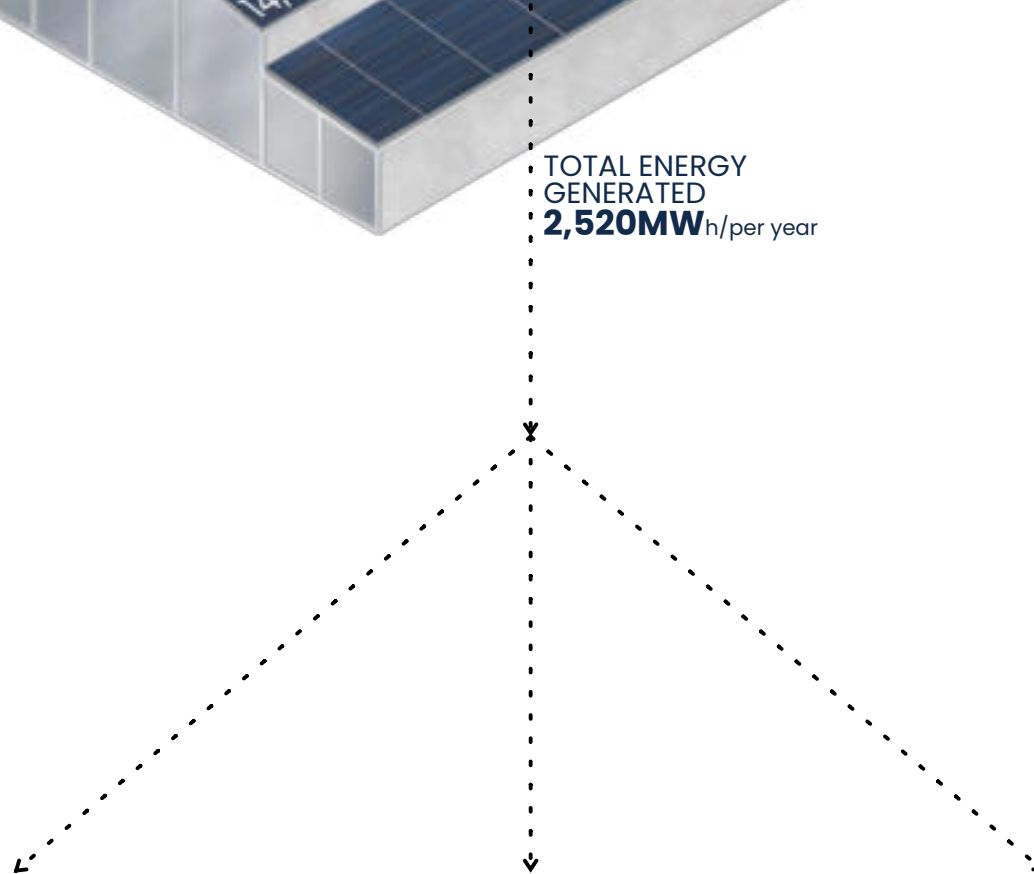
Adding photovoltaic panels to a roof lets energy be generated passively, and that energy can be fed back into the city's energy grid and used for site maintenance, such as lighting or staff accommodations.

If any energy is left over, it can be stored in storage and later sold to the city when prices are high, generating additional profit.

Electric vehicle charging is a profitable option, making this a point of interest for the city, as well as dedicating parking spots for residents to generate additional profit.



TOTAL ENERGY
GENERATED
2,520MW_h/per year



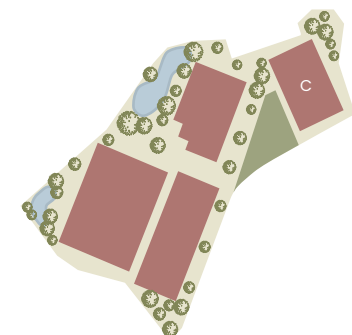
ENERGY
STORED
1,590 MW_h/per year



ELECTRIC
VEHICLE
CHARGING
70 MW_h/per year



SITE
MAINTENANCE
860MW_h/per year



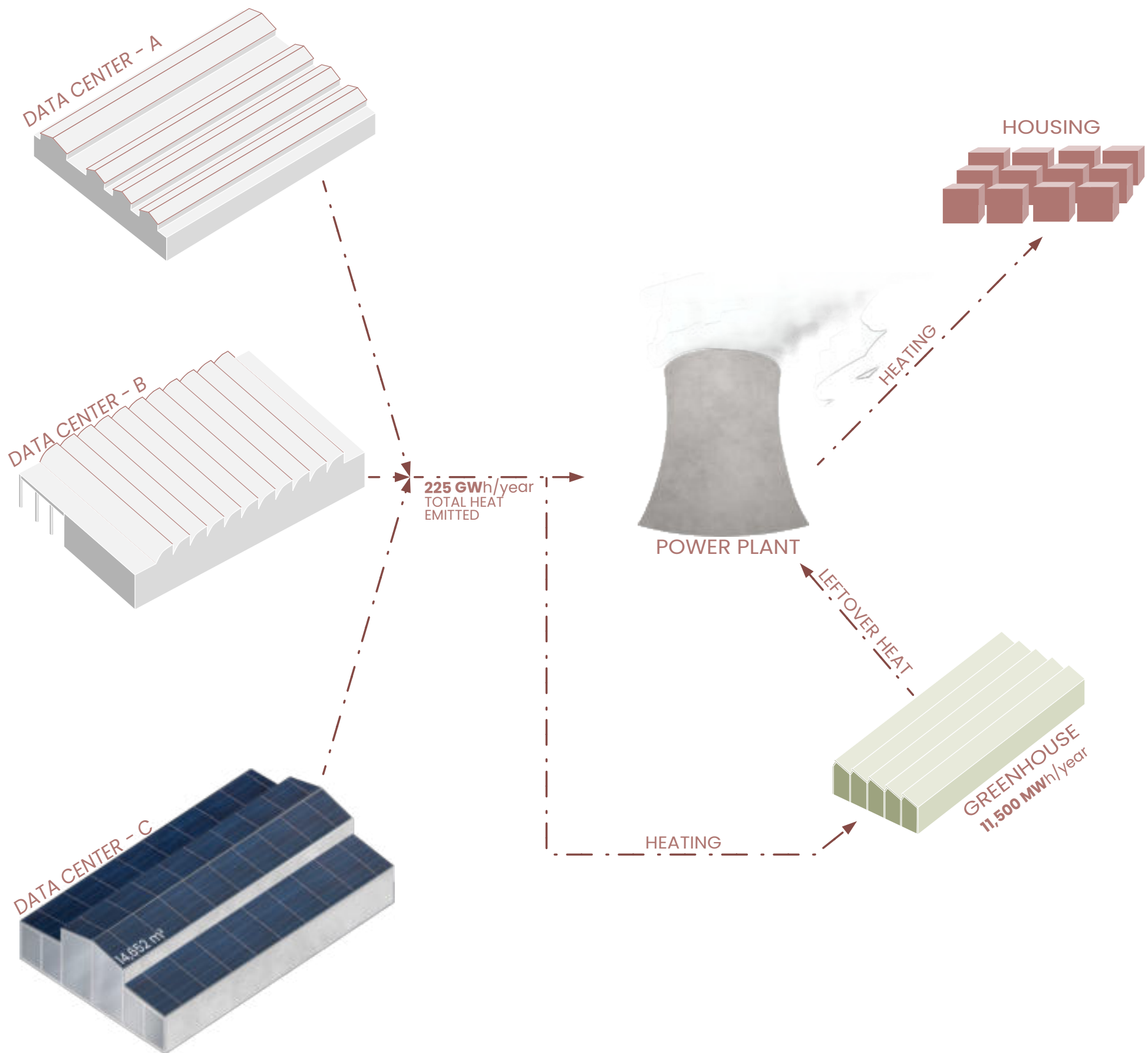
WASTE HEAT MANAGEMENT:

Excessive heat is a major problem for data centres due to the sheer amount of heat they emit throughout the year. To deal with this problem, I set up a greenhouse to grow vegetables year-round using the heat.

In the summer, the heat will travel through the power plant connected to the residential areas. There, the heat will be chilled and will provide cooling to residents through already existing infrastructure.

During the winter, the process is the same, but the heat is not cooled and sent to residential houses right away.

This arrangement will make the city more likely to support the project and generate additional profit with minimal effort.



FINAL STATEMENT:

During this experience, I developed strong time-management skills, learning to prioritize tasks and meet deadlines while still producing high-quality work. I had the opportunity to receive feedback from experienced professionals, whose advice helped me improve my thinking and approach. I was also lucky to work alongside people from a wide range of fields, each bringing knowledge and real-world experience that expanded my perspective

I got to see data centres in a new light and to understand the different typologies.

I got to reimagine them without real-world constraints like finances or legality, and to freely explore possible designs and solutions with supportive guidance.